

Analysis of Materials Science and Biological Databases

This document provides an assessment of selected databases based on their accessibility via Python packages, their underlying data format, and their cost model.

Materials Science Databases

MatWeb

Python Package Availability: No official or community-developed Python package for programmatic access.

PDF-Based Resource: No. Primary format is a web interface. Registered users can manually save individual data sheets as PDF or Excel files. Premium members can export data in XML format for CAD/FEA software.

Cost Model: Freemium. Searching and viewing data sheets is free. Advanced features require free registration or paid premium membership (\$99.95/year). Full programmatic access only available through high-cost corporate "Database Licensing" service.

Notes: Extensive database of manufacturer-supplied material properties, designed primarily as a web-based lookup tool for engineers. Not suitable for large-scale computational research or data mining in academic settings due to access model limitations.

Materials Project

Python Package Availability: Yes. Supported by two key packages:

- **pymatgen** - Powerful materials analysis library
- **mp-api** - Recommended client for database API interaction

PDF-Based Resource: No. Data is computationally generated and accessed programmatically via REST API returning JSON format. Python clients parse data into easy-to-use objects.

Cost Model: Free. All data and API access are free for all users. Registration

required to obtain a personal API key.

Notes: Gold standard for open-access computational materials data. API-first design makes it essential for materials discovery, screening, and machine learning applications. Highly recommended resource.

HydrogelDB

Python Package Availability: No. Packages with similar names on PyPI are unrelated tools.

PDF-Based Resource: No.

Cost Model: Not applicable.

Notes: Represents a conceptual need rather than an existing database. Field currently lacks centralized, API-accessible resources. Data is fragmented across scientific literature, often available as supplementary files requiring manual discovery and curation.

PolyInfo

Python Package Availability: No official package. Some third-party scrapers developed by community but may violate terms of service.

PDF-Based Resource: No. Data stored in relational database, available programmatically via REST API (JSON format) and SPARQL endpoint (RDF format).

Cost Model: Free, but requires user registration.

Notes: Valuable, professionally curated database of experimental polymer data sourced from scientific literature. Utility for large-scale data science significantly limited by terms of use, which explicitly prohibit mass data downloading and web scraping.

BioMatDB

Python Package Availability: No.

PDF-Based Resource: Not applicable.

Cost Model: Not applicable.

Notes: Not currently available. Ongoing EU-funded project to create future biomaterials database and marketplace. Project in development and prototyping phase, with scheduled completion in late 2024. Should be monitored for future availability.

MatPortal

Python Package Availability: No specific package for MatPortal. However, as an ontology repository built on standard OntoPortal platform, it can be accessed programmatically using generic Python tools like Ontology Access Kit (OAK).

PDF-Based Resource: No. Stores and serves ontologies (formal data schemas) in standard formats like OWL via REST API.

Cost Model: Free.

Notes: Not a database of material properties. Purpose is to store vocabularies and metadata frameworks used to describe materials data consistently. Tool for advanced data management, supporting FAIR (Findable, Accessible, Interoperable, Reusable) data principles.

Biological and Biomedical Databases

Cell Behavior Database

Python Package Availability: Yes. Most relevant tool is the **cellphonedb** Python package.

PDF-Based Resource: No. CellphoneDB is a software tool, not a static database. Takes user-provided data as input, typically in standard bioinformatics formats like TSV files.

Cost Model: Free and open-source.

Notes: Refers to bioinformatics tool CellphoneDB. This is an analytical tool used to infer potential cell-cell communication networks from single-cell gene expression data. Not a database of observed, measured cell behaviors like migration speed or adhesion force data.

3D Bioprinting Data Hub

Python Package Availability: No.

PDF-Based Resource: No.

Cost Model: Not applicable.

Notes: Emerging field currently lacking centralized, queryable data hub. Data landscape highly fragmented, consisting of non-standardized datasets published within individual research papers. Initiatives like CECT Bioink Database exist but are web portals for manual searching without API access. Work in this area requires significant manual data mining from literature.

Chemical and Molecular Databases

PubChem

Python Package Availability: Yes. **PubChemPy** package is a mature and widely used Python wrapper providing simple interface to PubChem API.

PDF-Based Resource: No. Data available through powerful REST API in various machine-readable formats, including JSON, XML, and SDF.

Cost Model: Free. Entire database and all associated services publicly accessible at no cost.

Notes: One of the world's largest and most important free chemistry databases. Links chemical structures to biological activities, patents, and literature, making it an essential resource. PubChemPy library makes it extremely convenient for programmatic use.

ChEBI (Chemical Entities of Biological Interest)

Python Package Availability: Yes. The official Python library is **libChEBIpy**.

PDF-Based Resource: No. Data accessible via modern REST API and can be downloaded in various formats, including TSV, SDF, and ontology format OWL.

Cost Model: Free and open-access.

Notes: High-quality, manually curated database and ontology focused on small molecules of biological interest. Key strengths are reliability and formal ontological structure, which defines relationships between chemical entities (e.g., caffeine is a purine alkaloid). Ideal for tasks requiring high-confidence identifiers

and semantic queries.

Summary Table

Database	Python Package	API Access	Cost	Best Use Case
MatWeb	✗	Limited	Freemium	Manual property lookup
Materials Project	✓	✓	Free	Computational materials research
HydrogelDB	✗	✗	N/A	Conceptual need only
PolyInfo	✗	✓ (restricted)	Free	Polymer property reference
BioMatDB	✗	✗	N/A	Future resource (in development)
MatPortal	Partial	✓	Free	Ontology/metadata management
Cell Behavior Database	✓	N/A	Free	Cell communication analysis
3D Bioprinting Data Hub	✗	✗	N/A	Literature mining required
PubChem	✓	✓	Free	Chemical structure/activity data
ChEBI	✓	✓	Free	Biological molecule ontology